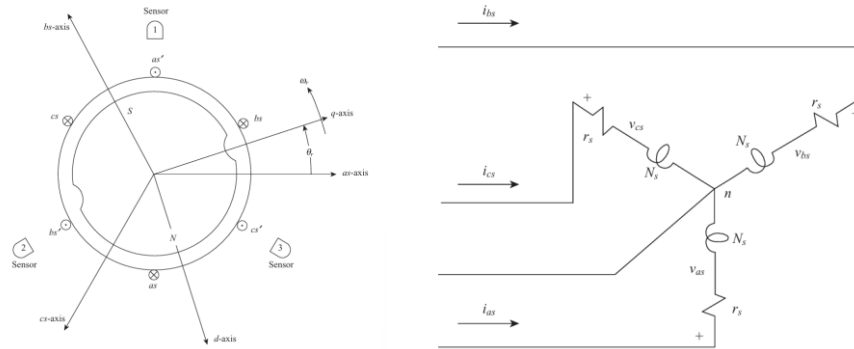


PM/Brushless DC Machines

Permanent magnet AC motors are widely recognized for their versatility and broad range of applications. By varying the complex control methods used to drive them, these motors can emulate torque characteristics typically associated with different motor types. This report focuses on the Permanent Magnet Brushless DC (PMBLDC) motor, named for its torque similarities to a conventional DC motor despite its distinct design and functionality. All the information presented in this report is sourced from the book Analysis of Electric Machinery and Drive Systems by Paul C. Krause, Scott D. Sudhoff, and Oleg Wasynczuk.

Operation Principle

The PMBLDC Motor is basically a three-phase permanent magnet synchronous machine. It consists of a rotor with permanent magnets and a stator with interconnected windings. The basic operating principle begins with the static magnetic field from the rotor's magnets interacting with the rotating magnetic field generated by the stator, which produces attraction/repulsion. This principle is the same as in PMAC motors, but in this case, the inverter configuration emulates the torque behavior of a brushed DC motor. In some cases, Hall effect sensors are used to provide feedback to the motor controller, helping to drive the signals accurately.



From these figures, we can observe a two-pole, three-phase permanent magnet machine. As mentioned, the rotor is equipped with permanent magnets and does not possess any windings. The stator, on the other hand, has three windings responsible for inducing the magnetic motive force (MMF). These windings are arranged 120 degrees apart mechanically. The stator windings are controlled by an electronic inverter, which supplies the currents in the appropriate sequence based on rotor position feedback. For a better understanding of the overall circuit and control system, refer to the next image depicting a PMAC machine setup, as it provides a clear representation of how the inverter interacts with the machine to regulate its operation.

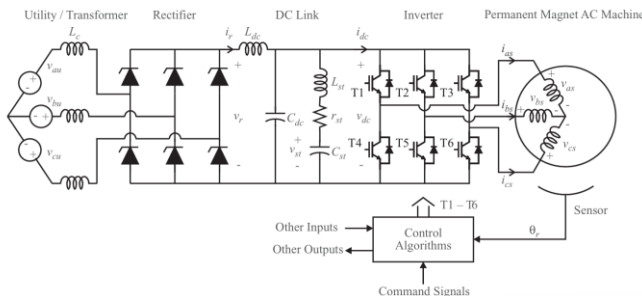


Table 8.6-1: Comparison of voltage and torque equations of a permanent-magnet dc machine and a brushless dc machine.

Permanent-magnet dc machine	Brushless dc machine
$V_a = r_a I_a + k_v \omega_r$	$V_{qs}^r = r_s I_{qs}^r + \frac{L_{ds}^r}{r_s} \omega_r^2 I_{qs}^r + \lambda_m^r \omega_r$
$T_e = k_t I_a$	$T_e = \frac{P}{2} \lambda_m^r I_{qs}^r$
$T_e = \frac{k_t V_a}{r_a} - \frac{k_v^2}{r_a} \omega_r$	$T_e = \frac{P}{2} \frac{r_s \lambda_m^r}{r_s^2 + \omega_r^2 L_{ds}^r} (\sqrt{2} V_s - \omega_r \lambda_m^r)$
$T_e(\text{stall}) = \frac{k_t V_a}{r_a}$	$T_e(\text{stall}) = \left(\frac{P}{2} \right) \frac{\lambda_m^r V_{gs}^r}{r_s}$
$\omega_r(T_e = 0) = \frac{V_a}{k_v}$	$\omega_r(T_e = 0) = \left(\frac{P}{2} \right) \frac{V_{gs}^r}{\lambda_m^r}$

Voltage and Torque Equations in Rotor Reference-Frame

$$\begin{aligned} (\lambda_{dq}^r)^T &= [\lambda_{ds}^r \quad -\lambda_{qs}^r \quad 0] \\ \mathbf{v}_{qd0s}^r &= \mathbf{r}_s \mathbf{i}_{qd0s}^r + \omega_r \boldsymbol{\lambda}_{dq0s}^r + p \boldsymbol{\lambda}_{qd0s}^r \\ \boldsymbol{\lambda}_{qd0s}^r &= \begin{bmatrix} L_{ls} + L_{mq} & 0 & 0 \\ 0 & L_{ls} + L_{md} & 0 \\ 0 & 0 & L_{ls} \end{bmatrix} \begin{bmatrix} i_{qs}^r \\ i_{ds}^r \\ i_{0s}^r \end{bmatrix} + \lambda_m^r \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} v_{qs}^r &= r_s i_{qs}^r + \omega_r \lambda_{ds}^r + p \lambda_{qs}^r \\ v_{ds}^r &= r_s i_{ds}^r - \omega_r \lambda_{qs}^r + p \lambda_{ds}^r \\ v_{0s}^r &= r_s i_{0s}^r + p \lambda_{0s}^r \end{aligned}$$

After deriving the voltage and torque equations in terms of machine variables, it's much easier to work with them in the rotor reference frame. Using the q, d, and 0-axis variables gives us a simpler way to model the motor, aligning everything with the rotor's magnetic field. With this approach, we can use the voltage and flux linkage equations to get a simplified torque equation, where the q-axis plays the main role in producing torque. This makes it easier to control and understand the motor's operation.

$$T_e = \left(\frac{3}{2}\right) \left(\frac{P}{2}\right) (\lambda_{ds}^r i_{qs}^r - \lambda_{qs}^r i_{ds}^r)$$

Analysis in Steady State Operation

In order to do a steady state analysis, we must consider the voltage equation in steady state thus the following $L_q = L_d$, $V_{qs}^r = \sqrt{2}V_s$, and $V_{ds}^r = 0$ ($\phi_v = 0$). This is because the inverter at these points maintains V_s to its peak. With this, and doing some algebra into V_{qs} we get that.

$$V_{qs}^r = \left(\frac{r_s^2 + \omega_r^2 L_q L_d}{r_s} \right) I_{qs}^r + \omega_r \lambda_m^r \quad \text{for } \phi_v = 0 \quad I_{ds}^r = \frac{\omega_r L_q}{r_s} I_{qs}^r \quad \text{for } \phi_v = 0$$

If we take and solve our previous equation for I_{qs} we can get the torque equation at steady state.

$$T_e = \left(\frac{3}{2}\right) \left(\frac{P}{2}\right) \frac{r_s \lambda_m^r}{r_s^2 + \omega_r^2 L_s^2} (V_{qs}^r - \omega_r \lambda_m^r) \quad \text{for } \phi_v = 0$$

From the two images below, we can see that the steady-state torque-speed graph exhibits a nearly linear relationship in the operational range, resembling the behavior of a traditional DC motor. However, as speed increases, the torque begins to drop due to the effects of back EMF and inductance. In contrast, the dynamic torque-speed graph reflects transient behaviors, such as fluctuations during acceleration or sudden load changes, which deviate from the steady-state characteristics. This highlights the differences between ideal steady-state operation and real-world performance influenced by external noise.

